TROPIC Simulation Model Analysis Report

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Current controlled release candidate: v0.3.0-clinical-simulation (unreleased) docs/RELEASE_NOTE_v0.3.0-clinical-simulation.md

Informative annex boundary. These deterministic simulation results are a data-free methods evaluation layered on the historical sealed clinical-simulation release. They are not MIDD evidence, a filing artifact, confirmatory efficacy evidence, sponsor approval, or evidence of regulator acceptance.

Reviewer Status Snapshot

Assessment	Status	Authoritative rationale	Meaning
Execution	PASS	Every run executed and requested/completed/failed accounting is complete.	Did requested replicates complete with failures explicitly accounted for?
Monte Carlo precision	PASS	Every scenario met its prespecified replication, failure-rate, and MCSE controls.	Were prespecified replication, failure-rate, and uncertainty criteria met?
Design operating characteristics	PASS	Analytic null, scenario-derived trial selection, and log-rank edge checks passed.	Did the design meet the scenario-specific scientific acceptance criteria?
Evidence qualification	NOT_QUALIFIED	Informational, data-free, non-MIDD, non-confirmatory methods evaluation only.	What decisions may these results support?

There is deliberately no single overall PASS. Execution and precision statuses describe the reliability of the computation; the design status describes the operating characteristics. Precision does not rescue an unacceptable design result.

Question, Context, and Model Risk

ICH M15 element	Assessed value
Question of interest	Under explicit public-aggregate or stress-test assumptions, what are the rejection probabilities and numerical precision of a one-sided unstratified log-rank test in a simplified TROPIC-like fixed two-arm OS or PFS design?
Context of use	Low-influence internal methods evaluation and reviewer training for simulation engineering, numerical precision, and sensitivity to stated design assumptions.
Model influence	LOW

ICH M15 element	Assessed value
Consequence of a wrong decision	Misinterpreting numerical operating characteristics as clinical evidence could misstate design performance or treatment benefit; the hard use boundary prevents those decisions from relying on this model.
Model risk	MODERATE
Model impact	LOW

Estimand and Intercurrent Events

ICH E9(R1) attribute	Implemented definition
Population	Hypothetical randomized participants meeting a simplified TROPIC-like analysis population; fixed public-randomized-design calibration of 377 control and 378 experimental participants. This is distinct from the repository's available real MP cohort of N=371 and does not imply that participant-level data are available.
Treatment conditions	control: Synthetic control strategy from randomization through follow-up.; experimental: Synthetic experimental strategy from randomization through follow-up.
Variables / endpoints	OS: Time from randomization to death from any cause.; PFS: Time from randomization to progression or death from any cause.
Population-level summary	Between-arm survival-distribution contrast evaluated by the one-sided intent-to-treat log-rank statistic; repeated-trial summary is rejection probability.

Intercurrent-Event Handling

Intercurrent event	Strategy	Simulation / analysis handling
permanent_treatment_discontinuation	TREATMENT_POLICY	Follow-up and endpoint ascertainment continue. Reference scenarios retain the assigned-arm hazard; stress scenarios explicitly allow effect waning after discontinuation without censoring the endpoint.
subsequent_anticancer_therapy	TREATMENT_POLICY	Endpoint follow-up would continue regardless of subsequent therapy; a separate therapy process is not generated in this bounded implementation.

Intercurrent event	Strategy	Simulation / analysis handling
death	OS: OUTCOME_EVENT_NOT_INTERCURRENT; PFS: COMPOSITE	Death is the OS event and is included in the PFS composite event.

Missing Data and Censoring

Mechanism	Classification	Handling
Independent Withdrawal	MISSING_ENDPOINT_OBSERVATION_NOT_INTERCURRENT_EVENT	Independently generated withdrawal censors at last observation. High-dropout scenarios stress this assumption; informative withdrawal is not modeled.
Administrative Censoring	Censoring	Censor at the fixed common analysis date after staggered enrollment.

Implemented ADEMP and OCTAVE Methods

Framework element	Implemented method
ADEMP aims	Estimate Type I error or power and its Monte Carlo uncertainty for the governed scenario grid.
ADEMP data-generating mechanism	Individual enrollment, piecewise-exponential endpoint time, independent withdrawal, optional treatment discontinuation with post-discontinuation waning, and fixed administrative censoring.
ADEMP methods	One-sided unstratified log-rank test at alpha 0.025.
ADEMP performance measures	Rejection numerator, denominator, probability, MCSE, and 95% Wilson interval.; Completed/failed accounting and failure reasons.; Mean event and censor proportions by randomized arm.
OCTAVE characteristics	OS/PFS baseline hazards, treatment-effect shape, withdrawal, and discontinuation assumptions.
OCTAVE trial design	Fixed parallel two-arm allocation with staggered enrollment and common analysis date.
OCTAVE analyses	Intent-to-treat one-sided log-rank analysis.
OCTAVE valuation metrics	Type I error or power, Monte Carlo precision, event/censoring rates, and execution failures.
OCTAVE evidence	Protocol/scenario/code hashes, deterministic trials, analytic benchmark, and machine-readable results.

Scenario Coverage

Scenario	Endpoint	Class	Assumption basis	Varying factors
OS_NULL_REFERENCE	OS	KEY_NULL	ASSUMED_REFERENCE	treatment hr segments=start month: 0; hazard ratio: 1; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0; post discontinuation hazard ratio=1
PFS_NULL_HIGH_DROPOUT	PFS	KEY_NULL_OPERATIONAL_STRESS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 1; withdrawal rate 12m=control: 0.25; experimental: 0.25; discontinuation rate 12m=0; post discontinuation hazard ratio=1
OS_PUBLISHED_EFFECT	OS	ALTERNATIVE	PUBLIC_AGGREGATE	treatment hr segments=start month: 0; hazard ratio: 0.7; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0; post discontinuation hazard ratio=1
PFS_PUBLISHED_EFFECT	PFS	ALTERNATIVE	PUBLIC_AGGREGATE	treatment hr segments=start month: 0; hazard ratio: 0.74; withdrawal rate 12m=control: 0.08; experimental: 0.08; discontinuation rate 12m=0; post discontinuation hazard ratio=1

Scenario	Endpoint	Class	Assumption basis	Varying factors
OS_MEDIAN_CALIBRATED	OS	ALTERNATIVE	PUBLIC_AGGREGATE_CALIBRATION	treatment hr segments=start month: 0; hazard ratio: 0.84106; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0; post discontinuation hazard ratio=1
PFS_MEDIAN_CALIBRATED	PFS	ALTERNATIVE	PUBLIC_AGGREGATE_CALIBRATION	treatment hr segments=start month: 0; hazard ratio: 0.5; withdrawal rate 12m=control: 0.08; experimental: 0.08; discontinuation rate 12m=0; post discontinuation hazard ratio=1
OS_DELAYED_NON_PH	OS	ALTERNATIVE_STRESS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 1; start month: 4; hazard ratio: 0.65; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0; post discontinuation hazard ratio=1
PFS_DELAYED_NON_PH	PFS	ALTERNATIVE_STRESS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 1; start month: 1; hazard ratio: 0.6; withdrawal rate 12m=control: 0.08; experimental: 0.08; discontinuation rate 12m=0; post discontinuation hazard ratio=1

Scenario	Endpoint	Class	Assumption basis	Varying factors
OS_WANING_AFTER_DISCONTINUATION	OS	ALTERNATIVE_STRESS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 0.7; withdrawal rate 12m=control: 0.04; experimental: 0.04; discontinuation rate 12m=0.35; post discontinuation hazard ratio=1
PFS_WANING_AFTER_DISCONTINUATION	PFS	ALTERNATIVE_STRESS	ASSUMED_STRESS	treatment hr segments=start month: 0; hazard ratio: 0.74; withdrawal rate 12m=control: 0.08; experimental: 0.08; discontinuation rate 12m=0.45; post discontinuation hazard ratio=1

The scenario registry is reported without pooling so reviewers can distinguish null, alternative, non-proportional-effect, and operational-stress behavior.

Exact Execution and Failure Accounting

Scenario	Requested	Completed	Failed	Failure rate	Failure detail
OS_NULL_REFERENCE	100000	100000	0	0.000%	Not applicable
PFS_NULL_HIGH_DROPOUT	100000	100000	0	0.000%	Not applicable
OS_PUBLISHED_EFFECT	25000	25000	0	0.000%	Not applicable
PFS_PUBLISHED_EFFECT	25000	25000	0	0.000%	Not applicable
OS_MEDIAN_CALIBRATED	25000	25000	0	0.000%	Not applicable
PFS_MEDIAN_CALIBRATED	25000	25000	0	0.000%	Not applicable
OS_DELAYED_NON_PH	25000	25000	0	0.000%	Not applicable

Scenario	Requested	Completed	Failed	Failure rate	Failure detail
PFS_DELAYED_NON_PH	25000	25000	0	0.000%	Not applicable
OS_WANING_AFTER_DISCONTINUATION	25000	25000	0	0.000%	Not applicable
PFS_WANING_AFTER_DISCONTINUATION	25000	25000	0	0.000%	Not applicable

Aggregate requested	Aggregate completed	Aggregate failed	Reconciliation
400000	400000	0	PASS

Execution Status Rationale

Scenario	Execution status	Execution rationale
OS_NULL_REFERENCE	PASS	Run executed and all replicates are accounted: requested=100000, completed=100000, failed=0.
PFS_NULL_HIGH_DROPOUT	PASS	Run executed and all replicates are accounted: requested=100000, completed=100000, failed=0.
OS_PUBLISHED_EFFECT	PASS	Run executed and all replicates are accounted: requested=25000, completed=25000, failed=0.
PFS_PUBLISHED_EFFECT	PASS	Run executed and all replicates are accounted: requested=25000, completed=25000, failed=0.
OS_MEDIAN_CALIBRATED	PASS	Run executed and all replicates are accounted: requested=25000, completed=25000, failed=0.
PFS_MEDIAN_CALIBRATED	PASS	Run executed and all replicates are accounted: requested=25000, completed=25000, failed=0.
OS_DELAYED_NON_PH	PASS	Run executed and all replicates are accounted: requested=25000, completed=25000, failed=0.
PFS_DELAYED_NON_PH	PASS	Run executed and all replicates are accounted: requested=25000, completed=25000, failed=0.

Scenario	Execution status	Execution rationale
OS_WANING_AFTER_DISCONTINUATION	PASS	Run executed and all replicates are accounted: requested=25000, completed=25000, failed=0.
PFS_WANING_AFTER_DISCONTINUATION	PASS	Run executed and all replicates are accounted: requested=25000, completed=25000, failed=0.

Every failed replicate remains in the denominator accounting above; failures are not silently deleted from the evidence surface.

Operating Characteristics and Monte Carlo Precision

Scenario	Class	Rejections / analyzed	Estimate	MCSE	Wilson 95% interval
OS_NULL_REFERENCE	KEY_NULL	2525 / 100000	2.525%	0.000496109	2.430% to 2.624%
PFS_NULL_HIGH_DROPOUT	KEY_NULL_OPERATIONAL_STRESS	2504 / 100000	2.504%	0.000494095	2.409% to 2.603%
OS_PUBLISHED_EFFECT	ALTERNATIVE	23688 / 25000	94.752%	0.00141033	94.469% to 95.022%
PFS_PUBLISHED_EFFECT	ALTERNATIVE	24581 / 25000	98.324%	0.000811889	98.157% to 98.476%
OS_MEDIAN_CALIBRATED	ALTERNATIVE	10919 / 25000	43.676%	0.00313688	43.062% to 44.292%
PFS_MEDIAN_CALIBRATED	ALTERNATIVE	25000 / 25000	100.000%	0	99.985% to 100.000%
OS_DELAYED_NON_PH	ALTERNATIVE_STRESS	20166 / 25000	80.664%	0.00249777	80.170% to 81.149%
PFS_DELAYED_NON_PH	ALTERNATIVE_STRESS	24564 / 25000	98.256%	0.000827909	98.086% to 98.411%
OS_WANING_AFTER_DISCONTINUATION	ALTERNATIVE_STRESS	19647 / 25000	78.588%	0.0025944	78.075% to 79.092%
PFS_WANING_AFTER_DISCONTINUATION	ALTERNATIVE_STRESS	24036 / 25000	96.144%	0.00121775	95.898% to 96.376%

Precision and Design Decisions

Scenario	Precision status	Precision rationale	Design status	Design rationale
OS_NULL_REFERENCE	PASS	Completed 100000 >= 100000; failure rate 0 <= 0.001; MCSE 0.000496109 <= 0.0005.	PASS	Wilson contains alpha=True; absolute deviation 0.00025 <= tolerance 0.00148833 is True.
PFS_NULL_HIGH_DROPOUT	PASS	Completed 100000 >= 100000; failure rate 0 <= 0.001; MCSE 0.000494095 <= 0.0005.	PASS	Wilson contains alpha=True; absolute deviation 4e-05 <= tolerance 0.00148229 is True.
OS_PUBLISHED_EFFECT	PASS	Completed 25000 >= 25000; failure rate 0 <= 0.001.	NOT_PREDEFINED	No minimum power criterion was prespecified; estimate is descriptive.
PFS_PUBLISHED_EFFECT	PASS	Completed 25000 >= 25000; failure rate 0 <= 0.001.	NOT_PREDEFINED	No minimum power criterion was prespecified; estimate is descriptive.
OS_MEDIAN_CALIBRATED	PASS	Completed 25000 >= 25000; failure rate 0 <= 0.001.	NOT_PREDEFINED	No minimum power criterion was prespecified; estimate is descriptive.
PFS_MEDIAN_CALIBRATED	PASS	Completed 25000 >= 25000; failure rate 0 <= 0.001.	NOT_PREDEFINED	No minimum power criterion was prespecified; estimate is descriptive.
OS_DELAYED_NON_PH	PASS	Completed 25000 >= 25000; failure rate 0 <= 0.001.	NOT_PREDEFINED	No minimum power criterion was prespecified; estimate is descriptive.
PFS_DELAYED_NON_PH	PASS	Completed 25000 >= 25000; failure rate 0 <= 0.001.	NOT_PREDEFINED	No minimum power criterion was prespecified; estimate is descriptive.
OS_WANING_AFTER_DISCONTINUATION	PASS	Completed 25000 >= 25000; failure rate 0 <= 0.001.	NOT_PREDEFINED	No minimum power criterion was prespecified; estimate is descriptive.

Scenario	Precision status	Precision rationale	Design status	Design rationale
PFS_WANING_AFTER_DISCONTINUATION	PASS	Completed 25000 >= 25000; failure rate 0 <= 0.001.	NOT_PREDEFINED	No minimum power criterion was prespecified; estimate is descriptive.

Probability estimates, numerators, denominators, Monte Carlo standard errors, and Wilson intervals above are rendered directly from the authoritative scientific JSON. No result number is manually transcribed into this report.

Prespecified Criteria and Interpretation

Control	Criterion	Scope
Key Null Min Completed	100000	All applicable scenarios
Alternative Min Completed	25000	All applicable scenarios
Alternative Precision Justification	At 25,000 completed replicates the worst-case binomial MCSE is no greater than sqrt(0.25/25000)=0.003163. Alternative and stress-scenario probabilities are descriptive because no sponsor-approved performance threshold or MCID is available.	All applicable scenarios
Target Mcse Key Null	0.0005	All applicable scenarios
Max Failure Rate	0.001	All applicable scenarios
Analytic Null Probability	0.025	All applicable scenarios
Analytic Null Abs Tolerance Floor	0.001	All applicable scenarios
Analytic Null Mcse Multiplier	3	All applicable scenarios
Wilson Confidence Level	0.95	All applicable scenarios
Null Acceptance	Wilson interval contains alpha and absolute deviation is no greater than max(0.001, 3MCSE); upper Wilson bound is not required to be below alpha.	All applicable scenarios

Design conclusions are scenario-specific. Null scenarios assess error control; alternatives and stress cases characterize behavior under their stated assumptions. Published effects, reconstructed comparators, or calibration values are not promoted to clinically justified target effects or minimum clinically important differences.

Representative Simulated Trial Paths

Role	Scenario	Search index	Seed binding	Events	Censors	Z statistic	One-sided p-value	Decision and selection evidence
reject	OS_NULL_REFERENCE	6	scenario=2026081401; selection=2026081490	control=235; experimental=211	control=142; experimental=167	-2.30637	0.0105451	REJECT -- First eligible trial in governed search order
non_reject	OS_NULL_REFERENCE	1	scenario=2026081401; selection=2026081490	control=232; experimental=235	control=145; experimental=143	0.0638509	0.525456	DO_NOT_REJECT -- First eligible trial in governed search order
near_alpha_boundary	OS_NULL_REFERENCE	2945	scenario=2026081401; selection=2026081490	control=232; experimental=211	control=145; experimental=167	-1.96066	0.0249594	REJECT -- absolute distance from alpha=4.05688e-05

These are actual aggregate paths selected from the governed OS null scenario using the frozen scenario binding, independent selection seed, and deterministic search rules. They are simulated trials, not original TROPIC participants, and they do not substitute for aggregate operating-characteristic estimates.

Deterministic Edge-Case Verification Fixtures

Fixture	Role	Seed	Analysis status	Failure reason	Z statistic	One-sided p-value	Reject	Events	Expected	Validation status
POSITIVE_FAVORABLE	positive	2026081491	COMPLETED	Not applicable	-7.51381	2.87157e-14	Yes	48	COMPLETED_REJECT	PASS
NEGATIVE_UNFAVORABLE	negative	2026081492	COMPLETED	Not applicable	7.51381	1	No	48	COMPLETED_DO_NOT_REJECT	PASS

Fixture	Role	Seed	Analysis status	Failure reason	Z statistic	One-sided p-value	Reject	Events	Expected	Validation status
BOUNDARY_NO_EVENTS	boundary_non_estimable	2026081493	FAILED	zero_logrank_variance	Not applicable -- non-estimable	Not applicable -- non-estimable	Not applicable -- non-estimable	0	FAILED_ZERO_VARIANCE	PASS

These deliberately artificial separated-time and zero-event fixtures verify sign, decision, and non-estimable boundary handling. They are software-validation cases, not representative simulated trial paths and not scientific operating-characteristic results.

Change Control and Deviations

Item	Final result
Deviations from the frozen MAP	None

Verification and Validation Evidence

Check	Evidence / result
Analytic Null Benchmark	status: PASS; rule: Wilson interval contains alpha and absolute deviation is no greater than max(0.001, 3MCSE); upper Wilson bound is not required to be below alpha.; scenario ids: OS_NULL_REFERENCE; PFS_NULL_HIGH_DROPOUT
Representative Trial Selection	status: PASS; roles found: reject; non_reject; near_alpha_boundary; note: Scenario-derived aggregate trial examples; no subject rows are retained.
Logrank Edge Fixture Status	status: PASS; note: Artificial positive, negative, and non-estimable fixtures; not simulated trial paths.
Accounting Identity	status: PASS; rule: requested = completed + failed for every scenario

Seeds, Hashes, and Reproduction

Observed Execution Environment

Component	Recorded identity
Python	3.11.15
NumPy	2.2.6
PyYAML	6.0.3
Floating-point dtype	float64
Random-number generator	numpy.random.PCG64
Dependency lock	requirements-ci.lock

Scenario Seed Ledger

Scenario	Seed	Scenario SHA-256
OS_NULL_REFERENCE	2026081401	7adc14f9810532d962c09ca7dbff26c177370ed9eed7e45ee1a5a23042d450a7
PFS_NULL_HIGH_DROPOUT	2026081402	8107f34d985c617abd9c505588fb5735f7f5602216420ec9e083024101c6183e
OS_PUBLISHED_EFFECT	2026081403	4aa624a9495fc3c5ea0f3fd41650e539618761281efcb710c71b13d68bd2f615
PFS_PUBLISHED_EFFECT	2026081404	abf18d3af342404d347c4f3efc140af77f51b988d4c2efdc13d31e8417b9edf5
OS_MEDIAN_CALIBRATED	2026081405	0b9c86fe0ad86e93cd56d2084cf4b5d31e775668efcae0aed188366e06bfaf70
PFS_MEDIAN_CALIBRATED	2026081406	61fc45f12ce1db4b691c82d8252f5263cc8ed225a5d2d482c9e985e1b5017028
OS_DELAYED_NON_PH	2026081407	8f4757706e1587997cc40bc36ecc5e7fe1a45a42744fe38c8be9a0eb8a2a10cb
PFS_DELAYED_NON_PH	2026081408	b857a9a989abae48810e5718eca46519c3bccdbba7929fa74a68bfce23b31727
OS_WANING_AFTER_DISCONTINUATION	2026081409	b24e9201196c430248db4ed62b7dd0b078d2afdd4f91a77682802c47c3e30374
PFS_WANING_AFTER_DISCONTINUATION	2026081410	ee4aae8c9cdb8db3d8898e754b3dcd55f4f1fd7f5646a0808b95cef13ccd33fd

Artifact Bindings

Artifact	SHA-256
Governed protocol	09798cd52adedd742a10f39266c67df0d9fe40b4c693912e4962b47601446b61
Authoritative result file	61a109fd903067a217c5df427908d3877a0bf54605af405f9226a31dab9fdbb3
Protocol recorded by result	09798cd52adedd742a10f39266c67df0d9fe40b4c693912e4962b47601446b61
Scenario registry	da115050f9d3fb69202b7154814c0eb204852efbe99f9b6107e548419f3f7768
Simulation code	3f5de78d6b174def4a8b92e77d4cde7064f25477f4cb18be76a295cbe17cc3ac
Scientific output	4e3b819118be9fc4e011a74d2a1f58d1a6526b9f31c25758f7ea7605b6d56cfb

Reproduce the scientific JSON with python3 platform/simulation_precision.py; rebuild both reviewer documents with python3 platform/build_simulation_report.py. Identical governed inputs and seeds must reproduce byte-identical scientific content and reviewer reports.

Limitations and Qualification Boundary

- Classification: NON_MIDD_NON_CONFIRMATORY_DATA_FREE_METHODS_EVALUATION
- Model Influence: LOW
- Evidence Status: INFORMATIONAL_ONLY
- Authoritative Patient Data Used: No
- External Validation Completed: No

This evidence surface does not establish authoritative CbzP subject-level data, a clinically justified minimum effect, external model validation, independent organizational review, sponsor approval, or regulator alignment. It must not be used for clinical, labeling, filing, or patient-level decisions.

References and Governing Basis

- TROPIC simulation-precision research basis
- ICH M15: General Principles for Model-Informed Drug Development
- ICH E9(R1): Estimands and Sensitivity Analysis
- FDA: Adaptive Designs for Clinical Trials of Drugs and Biologics
- FDA: Complex Innovative Trial Design Meeting Program
- ADEMP framework
- OCTAVE framework